Rev	Change	Released By	Date	Revision Description
A	0021273501	Walker Li	2019/12/11	Initial release

Bare FPC Independent QCP&CPP

080-03130

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Bare FPC Independent QCP&CPP 1
1. Purpose 3
2. Scope 3
3. Responsibilities 3
4. References 3
5. Definitions 3
6. Policy Statement 3
7. Procedure 4
7.1. Opportunities of creation for independent QCP and CPP 4
• When start new project. 4
• Process change 4
7.2. Design /new requirement review 4
7.3. DFMEA 4
7.4. Review with suppliers 4
7.5. Create independent QCP & CPP 4
7.6. DFM team/ fab team final review 4
7.7. Communicate with GSM team 4
7.8. Publish through agile 5
7.9. Follow up for execution 5
Appendix A: Revision History 7
Example: Bare flex Independent QCP & CPP 8

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1. Purpose

This document provides the guideline for creation of bare flex independent QCP and CPP.

2. Scope

This document applies for new project or process changes which are not applicable for items/ contents in general QCP and CPP in 080-01409 and 080-03022
There should be maximum one independent QCP/CPP per project, shared across all vendors building that project.

3. Responsibilities

DFM : Review design and extract uniqueness of new product or changes. Review with PD for design proposal and finalize design.

Application SQE: Product realization ,define Process flow and reflecting process controls and requirements into independent QCP and CPP .

Site SQE : Follow up QCP&CPP execution , issue feedback

GSM: Follow up with supplier to be in place resource needed in terms of new CAPEX , human resource , capacity planning .

4. References

080-01409 CPP Apple bare FPC critical process parameter
080- 03022 QCP Apple bare FPC Quality Control Plan
080-02999 CQRA Apple continuous Quality Reliability Assurance

5. Definitions

CPP : Apple bare FPC critical process parameter
QCP : Apple bare FPC Quality Control Plan
DFMEA : Design Failure mode and Effects Analysis

6. Policy Statement

Independent QCP& CPP states compulsory requirements for quality and process capability related .If suppliers are not compliance to this requirement, supplier has to submit corrective action to Apple team .

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If supplier have noncompliance to independent QCP& CPP ,refere to document <Guideline for Risk assessment and action for violation at requirement for further action>.

7. Procedure

7.1. Opportunities of creation for independent QCP and CPP

- When start new project.
- Process change

7.2. Design /new requirement review

- DFM should take lead to review design if it is manufacturable an feasible for product realization , raise the proposal to optimize design
- Application SQE should take lead to review new requirement for product .

7.3. DFMEA

- DFM/application SQE should take lead to consolidate and review DFMEA to de-risk all the new requirements and processes

7.4. Review with suppliers

- Application SQE should review with supplier to collect data and feedback
- Application SQE should review and drive flex vendor for DFMEA to de-risk new requirement /new design

7.5. Create independent QCP & CPP

- Application SQE should extract information from DFMEA and design review to define process control and process flow.
- Application SQE should create QCP and CPP , this document focus on unique items which are not stated in general QCP and CPP.
- Applicate SQE can integrate QCP and CPP , see appendix for template example , the document should include IQC and In process control /OQC and CQRA and CPP .if there is item unique from general QCP and CPP, application SQE should highlight in this document.

7.6. DFM team/ fab team final review

- Application SQE should have final review with DFM/fab for more inputs to finalize document .

7.7. Communicate with GSM team

- Review with GSM team to understand requirement in term of resource ,new CAPEX needed , so that GSM team can work for readiness in suppliers' side .

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7.8. Publish through agile

- Application SQE should release document through agile and get document no.

7.9. Follow up for execution

- Application SQE should organize a review meeting with QCP/CPP by suppliers
- Site SQE should be invited .
- site SQE should follow each supplier sites for execution of independent QCP/CPP
- Suppliers should convert requirements into internal document .
- Site SQE should drive for FACA on noncompliance item or refer to guideline for Risk assessment and action for violation at requirement for further action

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Flow Chart	DRI	Activities
Review with suppliers ⇩	DFM/ Application SQE	DFM initiate new project SQE initiate new requirement
Design /process review ⇩	DFM/ Application SQE	Manufacturability and feasibility review
DFMEA review ⇩	Application SQE	Risk review and mitigation plan
Review with suppliers ⇩	PCB/FPC Vendor	hearing feedback Review supplier capability
Create independent QCP & CPP ⇩	Application SQE	Review general QCP/CPP and create unique QCP/CPP
DFM/Fab team final review ⇩	Application SQE	Review with DFM/fab for more inputs to finalize document .
Communicate with GSM ⇩	Application SQE/GSM	Review with GSM team to understand requirement in term of resource ,new CAPEX needed , so that GSM team can work for readiness in suppliers' side .
Publish through Agile ⇩	Application SQE	release official document through agile
Follow up for execution	GSM/ Site SQE	

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Appendix A: Revision History

Revision	ECO	Description
A	0021273501	Initial Release of document

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Example: Bare flex Independent QCP & CPP

Version1
August 26, 2019

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1. About this Document

1.1 Audience: the people who benefit from this guide are:

- Flex Vendor Bare flex Quality, Production and relative departments
- Apple GSM to plan capacity and SPOR selection

1.2 Related Document:

- Flex MCO
- Flex ODB++
- Bare Flex general CPP Apple Bare FPC Critical Process Parameter
- 080-2265-E
- BOM and individual component data sheet
- Flex General Cosmetic Criteria
- Apple Flex In Process Control Request
- OQC Sampling Plan
- Bare flex general QCP : Apple Bare FPC Quality Control Plan
- CQRA : Continuous Quality Reliability Assurance .

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1.2 Scope

This document applies standards to Dock (D42/43/ N104) flexes . It should be applied as the standard to be followed for flex vendor in quality control at IQC/IPQC/OQC/CQRA and CPP

1.3 Responsibility

This document is integrated QCP& CPP&CQRA which contains unique items only from general QCP/CPP.

And flex vendors must have their internal QCP and CPP respectively which include more items than the list provided below.

2. Quality Control Plan

2.1 IQC Control Plan:

Refer to general QCP (Apple Bare FPC Quality Control Plan) ,otherwise , specify as below for special control items

Process	Control item	Sample size	Frequency	Tool	Criteria	Control chart
Copper pating	quick via peeling	5 pcs/lot	FA	checking SOP	Separation mode : Good : Cu pillar , PI based peeled NG: Separation from plating interface .	
Punching	Mic hole quality	1panel/FA	FA/ LPA	visual inspection	no overhang burr is allowed	

2.2 IPQC control plan

Refer to general QCP (Apple Bare FPC Quality Control Plan) ,additionally , specify more items as below

Process	Control item	Sample size	Frequency	Tool	Criteria	Control chart
IPQC	quick via peeling	5pcs/ lot	lot	Peeling via peeling	Separation mode : Good : Cu pillar , PI based peeled , NG: Separation from plating interface .	N/A

2.3 Flex OQC

Refer to general QCP (Apple Bare FPC Quality Control Plan) ,otherwise , specify as below for special items

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2.4 Flex CQRA-bare flex

(Refer to the latest SOP for Apple bare flex CQRA - defined in ok2ship)

1. Reflow test : ok to skip pre-conditioning as Nikkan material is unable to meet that .
2. thermal stress/hot oil : acceptable to survive at 260 c
3. FIB /IM : monitoring, acceptable for nano-crac
4. Test coupon design (CQRA) should be submitted for apple approval before build started (refer to D4X design rule)
5. waive for b-Hast for Dupont HXC-1225 (failed for b-Hast in D4X)

2.5 Critical Process Parameters - Bare Flex (Follow Apple bare flex CPP , otherwise highlight as below)

1. Via Drilling process

- Machine type : UV laser (compulsory)
- Drilling scraps removal (2mm guide hole) : preferred by vacuum removal by machine instead of human

2. Copper plating

- Plating line (outlayer) : Via filling by VCP (compulsory)
- Plating line (inner layer) : R2R plating , preferred : via filling if have

3. Via AVI and AOM

- AVI (outlayer/inner layer) : 100%
- AOM (outlayer/inner layer) :10%
- Clean room level : 10K less preferred.

4. button pad imaging

- Exposure (outlayer) : Preferred LDI , or proper qualification if by CCD

5. Patterning

- Exposure (outlayer) : LDI
- Exposure (inner layer) : R2R CCD

6. Etching

- Etching line (outlayer) : preferred 2 fluid etching + vacuum
- Etching line (inner layer) : preferred 2 fluid etching + vacuum

7. De-cap

- use pico-laser to have v-cut on transition area .
- De-capping by machine .

8. Solder mask

- Exposure : LDI, Manual registration/CCD are not allowed.
- Curing : box oven is preferred , or proper qualification is conducted as benchmark of box oven for parameters setting.

9. Lamination

- Bonding sheet must be seasoning under dryer oven for 12 hours min back to room temperature. (follow data sheet)
- L2/L3 must be pre-treated with micro-etching and bake160c @2hours with N2 before lamination.
- All the L1 and L4 before IL layup must be baked for 2 hours at 160c/N2 to remove moisture .
- Full lamination condition : 180c at 1.5hours + post baking after de-capping at 180C 1.5 hours.
- Final lamination (SUS/PI) should be placed after Mic hole forming ???

10. Micro-etching before layup

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- micro-etching before outlayer lay up should be processed by panel form right before layup

11. Lay up / Lamination

- Roll cleaning : Mechanical Rolling cleaning should be in the process flow independently
- Rolling cleanness : proper qualification should be conducted to determine how many cleaning pass .

3. Others

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